

Case study on Ecosystems: The Great Barrier Reef

Group 1C:
Elias, Javier, Jinfeng, Ulia

Introduction

The Great Barrier reef is off the northeastern coast of Australia. It is the largest deposit of coral in the world, has at least 300 species of hard coral, and a variety of marine life including lobsters, crayfish, prawns, crabs, and a variety of fishes. It is also a major tourist attraction.



Biotic vs. Abiotic Factor

Biotic environment refers to factors that are living in an ecosystem, such as animals and plants. Abiotic environment refers to the factors that are not living, such as temperature and water availability. The temperature, an abiotic factor of the Great Barrier Reef, can severely affect the biotic factors of the Great Barrier Reef, such as fish and corals. The temperature of the water the corals live in can affect the health of the corals. When the water temperature is too high, corals expel the algae living in their tissues, causing them to be more susceptible to diseases and death. This is called coral bleaching. This could also affect the fishes that live in corals.

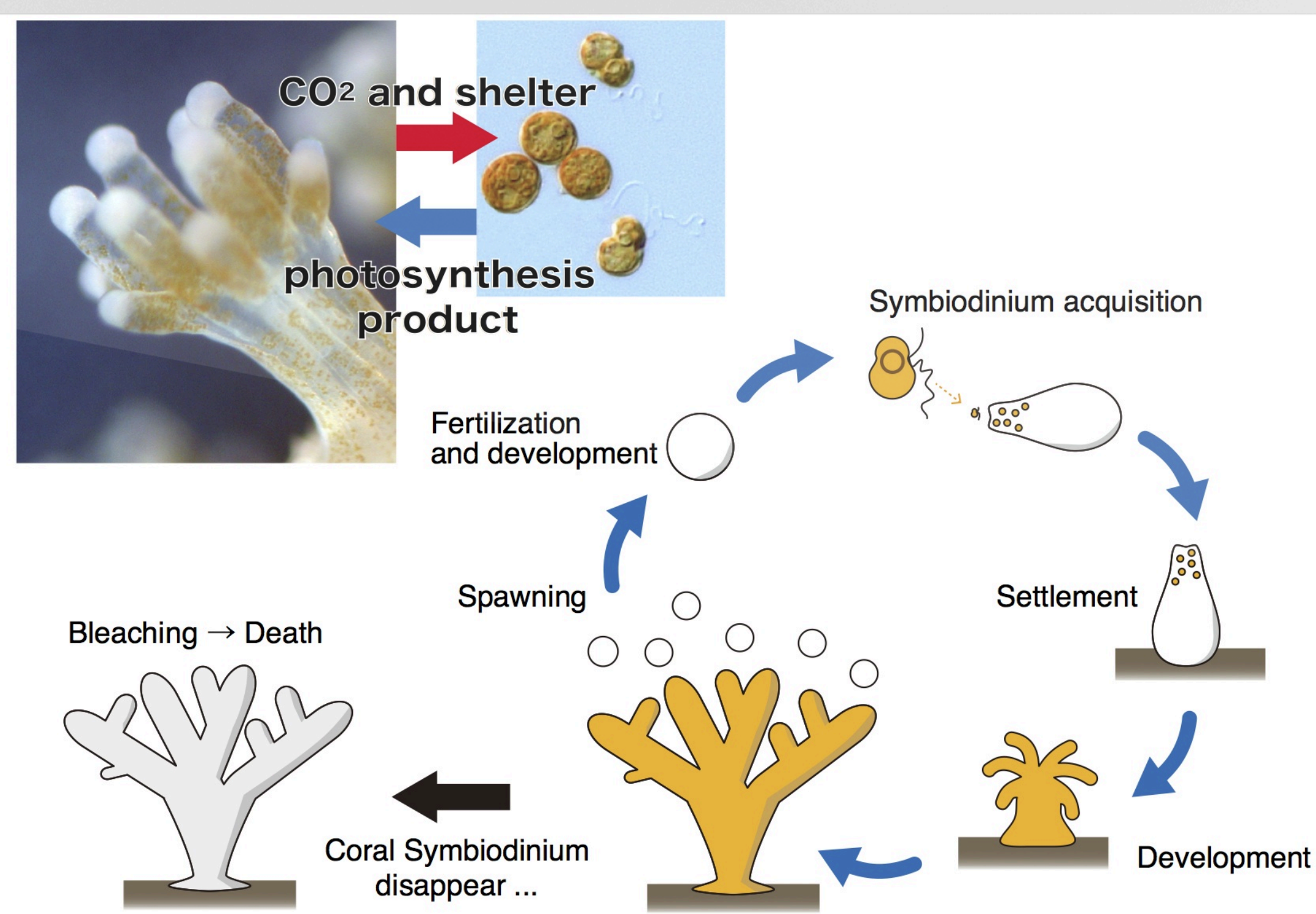
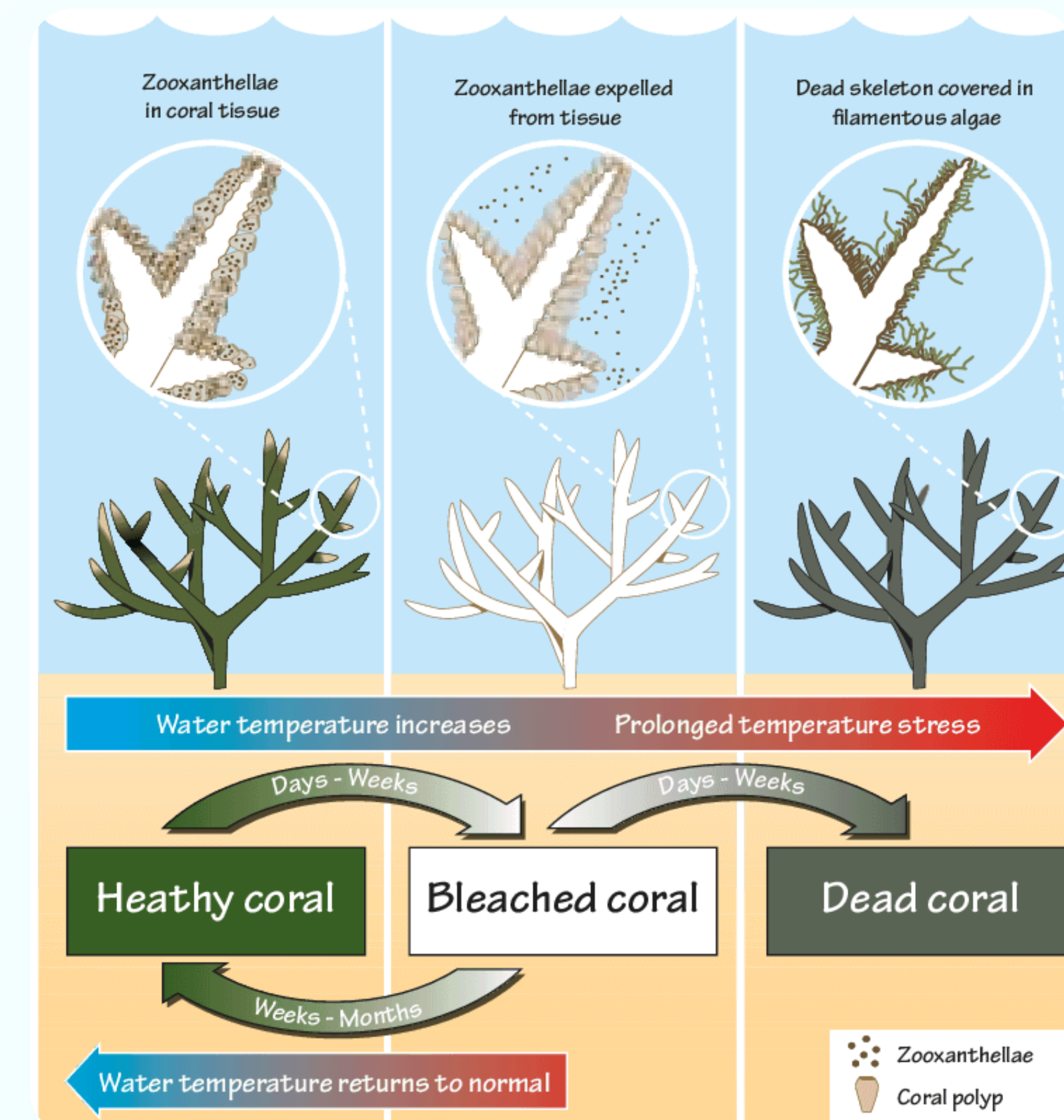


Figure 2. A symbiotic relationship between corals and Symbiodinium

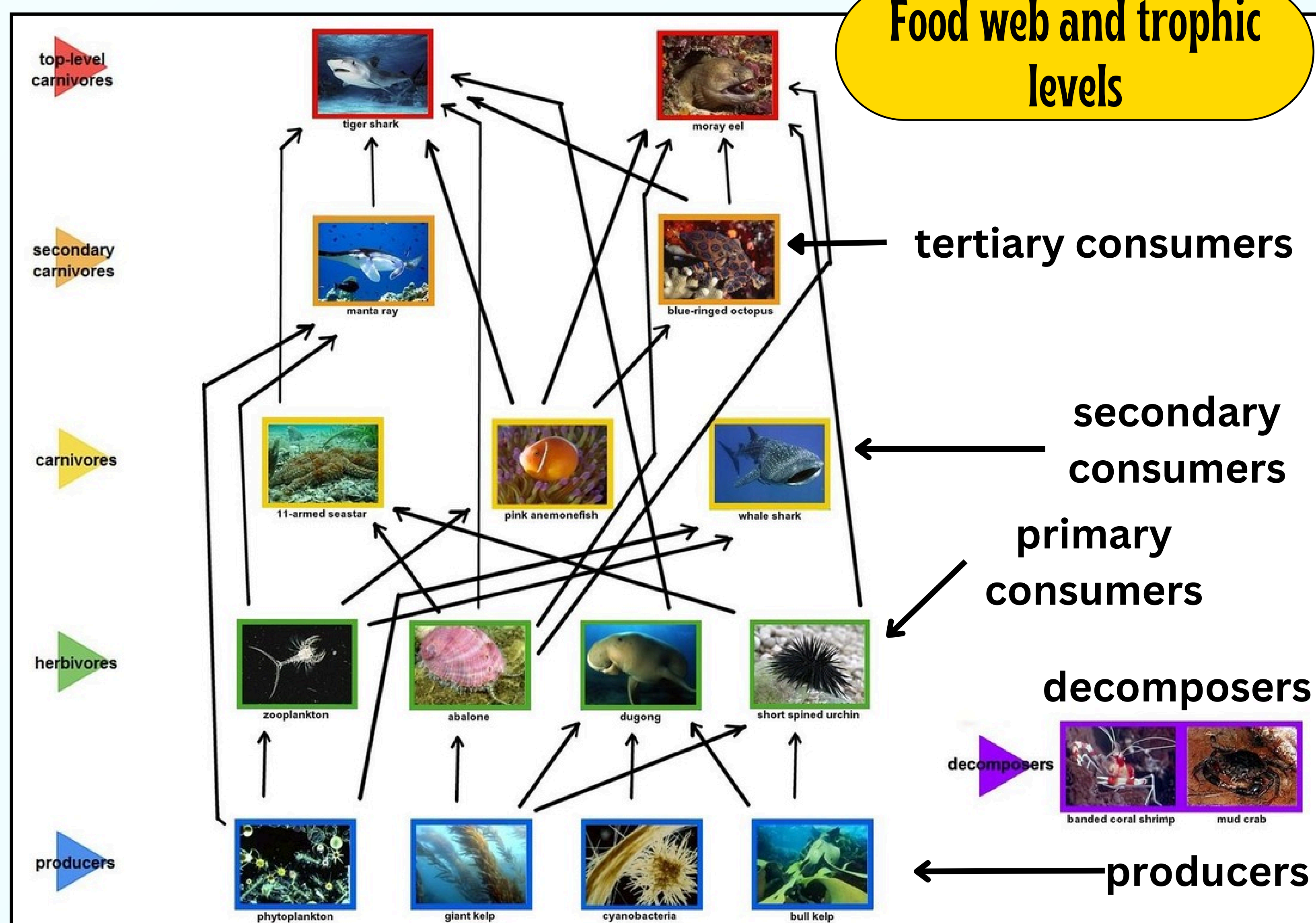
Case study of human impacts on coral reefs

Global warming is one of the biggest causes of the loss of coral reefs around the world. With oceans absorbing most of the excess heat on earth, the rapid increase in temperature can be detrimental to the coral reef ecosystems. When the temperature of the water stays warm for too long, corals expel the algae that helps produce food in them, causing them a lot of stress, which can cause death of the corals. With fishes depending on corals for shelter, the population of fishes could drop as a result of lesser corals. This could negatively affect the livelihood and food security of as many as half a billion people.

Types of interactions

Mutualistic relationships is a type of symbiotic relationship where all species benefit from each other. Corals have a mutualistic relationship with zooxanthellae, an alga. The algae produce nutrients such as glucose, glycerol, and amino acids as a product of photosynthesis, which the corals then use, to produce proteins, fats and carbohydrates. The coral in return provides a sheltered environment for them to photosynthesise.

Food web and trophic levels



Ecosystem conservation

To slow down the rate of global warming, we should tackle the **root cause**, reducing the amount of greenhouse gas emissions. There are multiple ways you can do this, some of which are either taking public transportation, using energy efficient appliances, or reducing the use of electrical appliances. We should also dispose of trash properly, as trash, which is marine debris, is harmful to coral reefs. Also, minimising fertiliser usage is essential to prevent water pollution, as excess runoff can harm water quality, damaging coral reefs.

Citations

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Environmental Protection Agency. (n.d.). What you can do to help protect coral reefs. EPA. <https://www.epa.gov/co-coral-reefs/what-you-can-do-help-protect-coral-reefs> - Ecosystem conservation

Images credit: 1st picture (<https://portdouglassmeridian.com/blog/10-fun-great-barrier-reef-facts/>) 2nd picture (<https://www.marineconservation.org.au/coral-bleaching/>)

3rd picture
(<https://www.dogonews.com/2015/12/4/can-lab-grown-super-coral-rescue-our-rapidly-dwindling-reefs/page/2/>)
4th picture
(<https://bckbiologywilliamson.weebly.com/biotic-factors.html>)